Individual Semester Project Report

Oliver Slater

Introduction

This report is a reflection on my personal contributions to the semester project and a summary of the processes I was involved in. I aim to reflect on the main challenges I encountered and overcame, as well as our strengths and weaknesses as a group.

Our project involved the design and programming of a car, encompassing multidisciplinary tasks in engineering, from mechanical design to electronics integration to coding. Our semester project completed a car prototype encompassing the mechanical design, 3D printing, and integration of an optocoupler sensor system for the measurement of speed.

I will outline my key contributions to the project: the design of the structure of the car, 3D printing of the essential components, and development of the speed measurement system. I will discuss some of the major challenges I faced in my work, such as the mechanical integrity of the design, optimization of the octocoupler system for the accurate calculation of speed, and some technical issues during assembly. I will also mention solutions put in place for resolving these problems and give insight into the interdisciplinary process that brought this project to completion.

The aim of this report is to showcase the progress, achievements, and areas that need further improvement for the future.

Project Background and Setup

The project was interdisciplinary and required contributions from all six of our group members. To increase efficiency within our group and ensure effective collaboration, we first developed a skills chart to highlight each member's strengths and areas of expertise. This helped us divide tasks in a way that maximized proficiency and utilized everyone’s abilities.

I primarily contributed to the mechanical design, 3D printing, and implementation of a speed measuring system using an octocoupler and Timer 2 configuration. Structuring our approach in this way enabled us to work on different aspects of the project in parallel while ensuring all members contributed meaningfully.

The project's major features included the car's mechanical structure design, fabrication of its parts through 3D printing, and integration of electronics for speed measurement and overall system performance. For it to function cohesively, its mechanical design, electronics, and software needed to be seamlessly integrated.

Speed Measurement System: Implementation and Code Development

I was also involved in the coding of the speed measurement logic, which involved using an optocoupler to detect rotational pulses for calculating the car's speed.

The first issue encountered during the early stages of this process was maintaining the car's speed consistently and precisely. I decided that since the necessary PWM value for the car to drive at a specific speed varies due to multiple factors such as gravity and car weight, the car must be able to actively adjust the PWM according to the current speed. Initially, the car's speed oscillated around a particular PWM value, making it impossible to maintain a constant velocity. This issue was in the approach I initially took for calculating the current speed.

Initially, I tried calculating speed by measuring the number of pulses per second generated by the optocoupler. However, this approach proved inadequate due to its low resolution at low speeds. To address this, I adopted the method of measuring the time interval between pulses, which provided significantly higher resolution and allowed for precise velocity control even at lower speeds. Configuring Timer 2 to count time intervals in millisecond units became a key aspect of the code. This system measured car speed by calculating the time between consecutive pulses and allowed for unit conversions, such as meters per second to kilometers per hour, for flexibility in system output.

Mechanical Design and Iterations

On the mechanical side of the project, we collaborated on creating custom components to house the car's systems securely and efficiently. Using measurements provided by Paval, including shaft dimensions, motor gears, and the clearance required for the optocoupler, I designed the brackets and base plate. These components were critical for ensuring proper alignment and secure mounting of all mechanical and electronic parts.

The design had some flaws which I discovered after the first print. The baseplate needed to be thicker and the tolerances had to be adjusted. Early versions of the brackets experienced stress-induced snapping due to the plastic material, necessitating redesigns to strengthen the structure. These issues were addressed through careful adjustments and testing of each design iteration.

Through this process, I gained a deeper understanding of mechanical tolerances and the importance of material strength in functional prototypes. The final designs successfully created a car capable of the project's requirements, providing a sturdy and precise framework for the car's components. This experience has taught me the advantages and disadvantages of a prototyping approach to design and the value of persistence in overcoming practical challenges.

Major Challenges and Lessons Learned

The most difficult challenge I faced in this project was coding the functionality to make the car consistently drive at the requested speed.

Debugging the speed control code was particularly sensitive, as the timing calculations and PWM adjustments were unstable. For instance, bugs in the speed measurement system caused erratic behavior, requiring troubleshooting to fine-tune the control loop and validate timing intervals for improved accuracy and reliability.

Beyond technical challenges, I also struggled with integrating the work distributed among group members. While task delegation allowed parallel progress, bringing together the various parts into a single design proved challenging.

These challenges were further exacerbated by group members being separated during holidays, which limited collaboration and real-time problem-solving. In hindsight, regular virtual meetings or shared work sessions could have mitigated these issues. Despite these difficulties, I learned valuable lessons about communication, testing, and collaboration in team projects.

Individual Report

Introduction

- I'm Ramal Abbasov, and I'm working on the semester project with group 8. Although I made an effort to contribute as much as I could to each assignment along the road, I spent most of my time working on the car's mechanical components. I created the Wheel Adapter in NX for challenge 1. Pavel and I coded the Arduino and motor driver connection in task 2. By the end, my attention was on the car's frame. After obtaining the frame measurements online, we 3D printed it, but there were still some issues throughout. To make the frame fit on the main base, I cut and altered it. Pavel and I also built, and 3D printed the holder for the frame. Overall, the car itself may not look the best, but every other part was done with as much effort as possible.

Reflection

- When I think back on my experience working with Group 8 on the semester project, I see how much I've learnt and how much work we all had to put in to make the project a reality. Even though my main concentration was on the car's mechanical parts, I made sure to give every work my all because I was committed to contribute significantly to every aspect of the project from the start. I was given the challenge of designing the Wheel Adapter in NX for Challenge 1. I was able to use my CAD design skills and gain a deeper understanding of how minor components affect a mechanical system's overall performance. As part of Task 2, Pavel and I worked together to code the motor driver and Arduino so the car could move and operate as planned. For me, the experience of coding was a novel challenge and a fantastic chance to learn more about this topic. As the project developed, my attention shifted to the car's frame. We found the required frame specifications online and proceeded to 3D print it, however we ran into some unforeseen problems. As an alternative to giving up, I modified and chopped the frame to suit the main foundation. Although this method called for perseverance and problem-solving abilities, I was pleased to see the finished frame. Pavel and I also collaborated to design and 3D print a holder for the frame, which added another level of complexity to the project. While the car's final appearance was not ideal, I felt that every aspect of it, from the mechanical components to the frame, had been completed with as much time and attention as I could provide.

Conclusion

- I feel a sense of accomplishment when I think back on the project because I was able to contribute in a variety of ways and persevered through difficulties. Even though there were some issues with the finished product, the process taught me a lot about problem-solving, teamwork, and the value of hard work at every stage of a project. In my future undertakings, I hope to put these lessons into practice and keep improving my mechanical design and teamwork abilities.

Individual Report Kiarash

Introduction:

The project involved developing an electric car capable of receiving user instructions and executing them accurately. The car was designed to run on two parallel rails, maintaining a straight path. It needed to calculate speed based on provided distance and time and perform accordingly. For this task, the car was programmed to travel 1 meter in 21 seconds, followed by 3 meters in 52 seconds, and then stop at the 4-meter mark. The project emphasized precision in calculations and performance, ensuring the car operated as intended within the given constraints and met the objectives of the assigned tasks.

Objectives:

In this project, my goal was to deepen my understanding of various aspects of our engineering program while developing essential skills for constructing an electric car using the provided materials. I aimed to enhance my technical knowledge and gain hands-on experience in designing and building such a vehicle.

Furthermore, I sought to improve my teamwork abilities, recognizing that collaboration can be challenging. My objective was to step out of my comfort zone, communicate more effectively with peers, foster stronger connections, and understand the dynamics of distributed work. This project served as an opportunity to grow both technically and personally.

Methodology:

To achieve my objectives for this project, I employed various methods and techniques. I actively participated in all stages of the project, from theoretical planning to practical implementation, ensuring I remained involved and contributed meaningfully. To enhance my teamwork skills, I made it a priority to communicate openly with my peers about their individual tasks and progress.

Several approaches proved particularly effective in fostering better collaboration and knowledge-sharing. Engaging with my teammates allowed me to learn new skills and perspectives, while my contributions helped strengthen our connections. This mutual exchange created a more cohesive and supportive team dynamic. Additionally, I prioritized providing constructive feedback and attentively listening to my teammates as they shared their work and creative ideas. This not only improved my understanding but also demonstrated my respect for their efforts. These methods collectively enabled me to grow both as an individual and as a team member.

Contribution:

After the first meeting, we divided the initial task of mechanical design. Each team member was assigned to design a specific case for the electrical components provided. My responsibility was to design an encoder wheel that would repeatedly block and unblock the sensor of the optocoupler. This task was particularly challenging, as it required multiple trials to ensure the encoder wheel's hole diameter fit snugly on the shaft without sliding. To accomplish this, I had to work with tight tolerances and refine my skills using the laser cutter machine, which was a rewarding learning experience.

During our second meeting, we tackled the programming phase, which included tasks such as programming the Nexion screen, optocoupler, Arduino, and motor driver. Each component had to be programmed individually and integrated into a single cohesive programming file. While my peers worked on these tasks, I helped and observed their progress, gaining valuable insights into the coding process. Simultaneously, I focused on assembling the cases for the components that had been designed and 3D-printed, ensuring they fit correctly and determining the best assembly methods. This step allowed us to identify potential adjustments and ensure everything was properly aligned.

In the final meeting, we assessed the remaining tasks and planned our timeline to complete the project. We decided on the car's frame design, discussed circuit handling, and allocated responsibilities for the holiday period. As I was the first to depart, I ensured all components had their cases securely bolted to the car's main base plate. Additionally, I checked upon the 3D printing of the car frame. This ensured that my tasks were finished, and my teammates could continue their work seamlessly during my absence.

Results:

For the mechanical task, I focused on designing the encoder wheel, gaining valuable experience in precision design and problem-solving. I also learned to use measurement devices like the oscilloscope and enhanced my skills in coding the Arduino and its connected components. Additionally, I took on a leadership role, ensuring the project progressed smoothly and stayed on schedule. Before my departure, I responsibly handed over my role to the next team member, ensuring continuity in leadership and project execution. This experience significantly contributed to my technical growth and my ability to manage and collaborate effectively within a team setting.

Reflection:

During this project, I gained valuable knowledge and developed a range of skills essential for engineering and teamwork. As the group manager for most of the project, my responsibilities included organizing meetings to discuss progress and tasks, ensuring each team member focused on their assigned duties, and keeping everyone on track to meet deadlines. While this leadership

role was exciting and fulfilling, it also presented challenges, particularly in coordinating with individuals who had different schedules, priorities, and work styles. These obstacles provided a valuable learning opportunity, pushing me to improve my communication skills and adapt to diverse perspectives within the team.

Initially, I believed building a car would be straightforward with the materials provided. However, the process proved far more complex than I anticipated. Engineers must address numerous factors when constructing even a single component of a car. This involves integrating theoretical knowledge, such as programming and circuit design, with hands-on tasks like assembling parts and ensuring precise fits through drilling and adjustments. This experience deepened my understanding of the intricate balance between theory and practice in engineering, highlighting the critical thinking and problem-solving required to bring designs to life. Ultimately, it was a challenging but highly rewarding journey of learning and growth.

SEMESTER PROJECT INDIVIDUAL REPORT- Iustinia-Vera Butescu

INTRODUCTION

Reflecting on the RC toy car project, I can say that it was a valuable learning experience. The project tested not only my technical skills but also my ability to adapt and problem-solve in real-time. I learned a lot about balancing different priorities.

SKILLS

Material Selection and Manufacturing Processes

This project taught me how important it is to choose the right materials for the car. We used 3D printing for most parts because it's fast and light, but we also saw that the plastic was not as strong as other materials. This helped us understand that it's important to balance strength and cost when choosing materials.

Problem Solving

I learned a lot about fixing problems during this project. I think this was the most important one, even though it was stressful because it was something new for all of us so we couldn't prevent the problems and it was also difficult to cooperate when we weren't sure what was the right thing to do.

Understanding Hardware-Software Interaction

I gained a better understanding of how hardware and software work together. Even though the final result was not what we hoped we had try to understand a whole new language and even thought I hadn't it master it , it was challenging and interesting to ask for others help and not just read information on a book: that was probably the most difficult part for me because I was used to learn by heart from notes and books ,while programming is more about trying and never know if it's going to be correct or wrong, if it is your fault or a system problem.

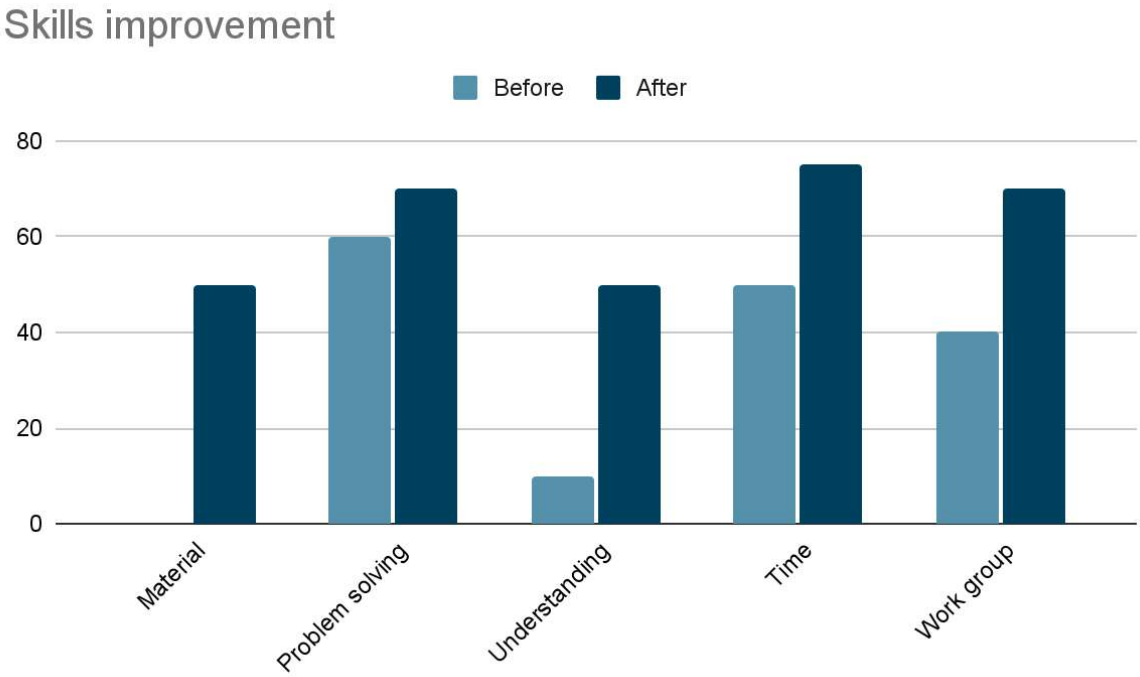
Time Management and Project Planning

This project also helped me learn how to manage my time(even though not completely). It is already difficult to work on things without wasting too much time on useless things or not thinking enough on others. But it is more difficult to do that when you depend on someone's work and they depend on yours.

Work group

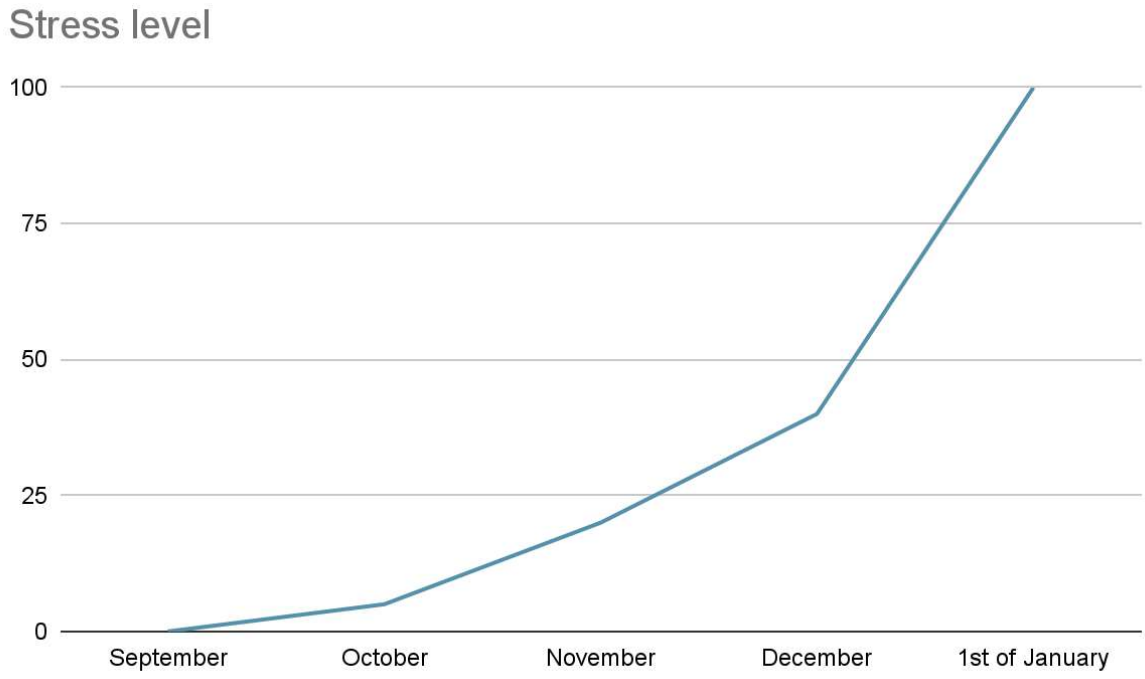
This last one is probably the most difficult for me, because even though I was used to doing school projects when I was in High school it was nothing like this. It was easy and

comforting to work with people I already knew, and we had more guidelines to follow and we already had all the information we needed and we knew what we were good at. While for the project I felt like I didn't understand anything entirely and since none of us was sure it was difficult sometimes to communicate and to understand what the others meant. But in the end I realized that it is important to say what we think , but also sometimes to step back and trust the others , even if we don't have the same opinion, because at the end it is important to hear all of them to have a complete view of the situation.



REFLECTIONS

Overall, I'm proud of the outcome, even though there were some issues with the programming and communication between the components. The project taught me valuable lessons in design, manufacturing, and programming, all while honing my ability to work with others and adapt to challenges.


Stress level
100
75
50
25
0
September
October
November
December
1st of January

Pavel Chladek

At the beginning of our project, I was afraid about how it would go. It was mostly because of new
People I had never seen before, so Il did not know how they would think, act and react.
Time by time I would say that we created well working team, and our discussions went smoothly
And almost every time we found any solution and compromise... at lease in my opinion.
I like that before we started to do anything, we decided who would do what so everything went
Nicely.
I took mechanical and electrical parts. For me in the electrical part everything was clearly
Readable from the materials we got, so we did not have any problems here.
On mechanical part I created a group with Kiarash and Ramal. Here l had time to get to know
Those guys. We were solving problems together and we talked a lot, not only about the project.
In summary I would say that it was a nice opportunity to meet other classmates and improve my
Talking and problem-solving skills.

Alessandra-Andreea Pricop

Introduction:

In order to properly make a remote-control (RC) car toy work you need to implement a coding system for the Arduino and the remote you are going to use for the toy car. As a customer you want the toy car working properly, doing what the description of the toy car says and also be able to change the speed and distance at any time as you want and as many times as you would like.

Problem statement:

The main problem when it comes to making the car start, move and stop when you command it to is having issues with the coding of the Arduino or even the Nextion Display while trying to keep the coding clean and not messy so other users can read through and understand what is going on.

Key criteria:

1. **Upload the code to the Arduino** – Should not run into any type of errors while uploading the full code to it.
2. **Connect the Arduino to the Nextion Display** – They should be able to communicate through UART communication without any issues.
3. **Nextion Display communication –** Should be able to communicate with the Arduino back and send the values we put in.
4. **Customer satisfaction** – Should run smoothly and make the customer happy.

Additional criteria:

1. **Pretty Nextion Display design –** Should be appealing to the customers.
2. **Match the car's design –** If the car is Lighting Mc'Queen, then the Nextion Display should include something of that sort.

Delimitation:

Some programming parts have been provided by the libraries for the Arduino Nano but also some files that I have found useful within the tutorial for the Nextion Display software in the Semester Project lectures. Additional coding was done by me with the help of either YouTube tutorials or older friends that guided me through this task.

Success Rate:

1. **Upload the code to the Arduino –** We did not run into any type of issues while uploading the code to the Arduino since the lines of code did not have any errors.

2. **Connect the Arduino to the Nextion Display –** We ran into some problems that we unfortunately couldn't figure out until the deadline and we were unable to connect the display to the Arduino properly so they would be able to connect through UART communication.
3. **Nextion Display communication –** We weren't able to achieve this key criteria due to issues either within the code or with the components we were given.
4. **Customer satisfaction –** We couldn't achieve the satisfaction of the customers due to not being able to connect the display to the Arduino.

Risk assesment:

A thorough risk assessment has been done to ensure that coding the Nextion Display and Arduino would ensure a success in order to have a working remote-control (RC) toy car until the deadline and to also be sure we would fulfill our customers satisfaction.

Design:

The design for the Nextion Display was mostly insipired by the movie "Cars" and the protagonist Lighting Mc'Queen. I thought it would be a good idea to implement some kind of connection between the two and so I added the background picture of him as our "Welcome" page.
The rest of the pages haven't been inspired by anything, the colors that I used were chosen because I like pink and shades of pink.

Coding the Arduino:

While coding the Arduino I first started by watching the tutorials from our Semester project and slowly trying to follow every line and practice it on my own Nextion Display in order to understand what was going on.

After finishing watching those I started on watching a few videos on YouTube about interrupts in codes and coding with a Nextion Display in order to be able to make the two communicate properly and be able to finish the project in time, also be able to fulfill my aspirations for this semester.

As someone who has never coded in their life it was a bit challenging at first and didn't really understand what was going on due to never doing this. I have asked for help from older students and from some of my friends who understood coding better than I did. Asking for help really helped me throughout the semester's project task and also helped me understand what was going on wrong with the code (sometimes).

I have organized and wrote what the code does in Visual Code to let other users know what the code does without needing to know a lot about coding. Organizing the code will just make everything easier and will provide a safety route if something goes wrong later in the code.

A few lines of code that have provided a great challenge was understanding the battery voltage and how to make the Arduino read it and send it to the Nextion Display if the battery voltage was

too low. It took me around 3 days to understand interrupts in code and a few more to understand how the code actually worked. I have asked for help from other older students to explain it to me to be able to code it properly so it would work.

Voltage code:

```
float read_adc(void)
{
  ADMUX = ADMUX | 0X40;
  ADCSRB = ADCSRB & (0XF8);
  ADCSRA = ADCSRA | 0XE7;
  unsigned int adclow = ADCL;
  return (((adclow + ((ADCH & 0x03) << 8))*5/1023)*5/3);
}
```

Nextion Display:

Working with the Nextion Display was a bit challenging at first since the semester project UART color pictures weren't matching with the ones we have now. After being able to figure out what color goes to what it was easy soldering it and connecting it to my laptop and successfully upload my Nextion Display design to the Nextion Display.

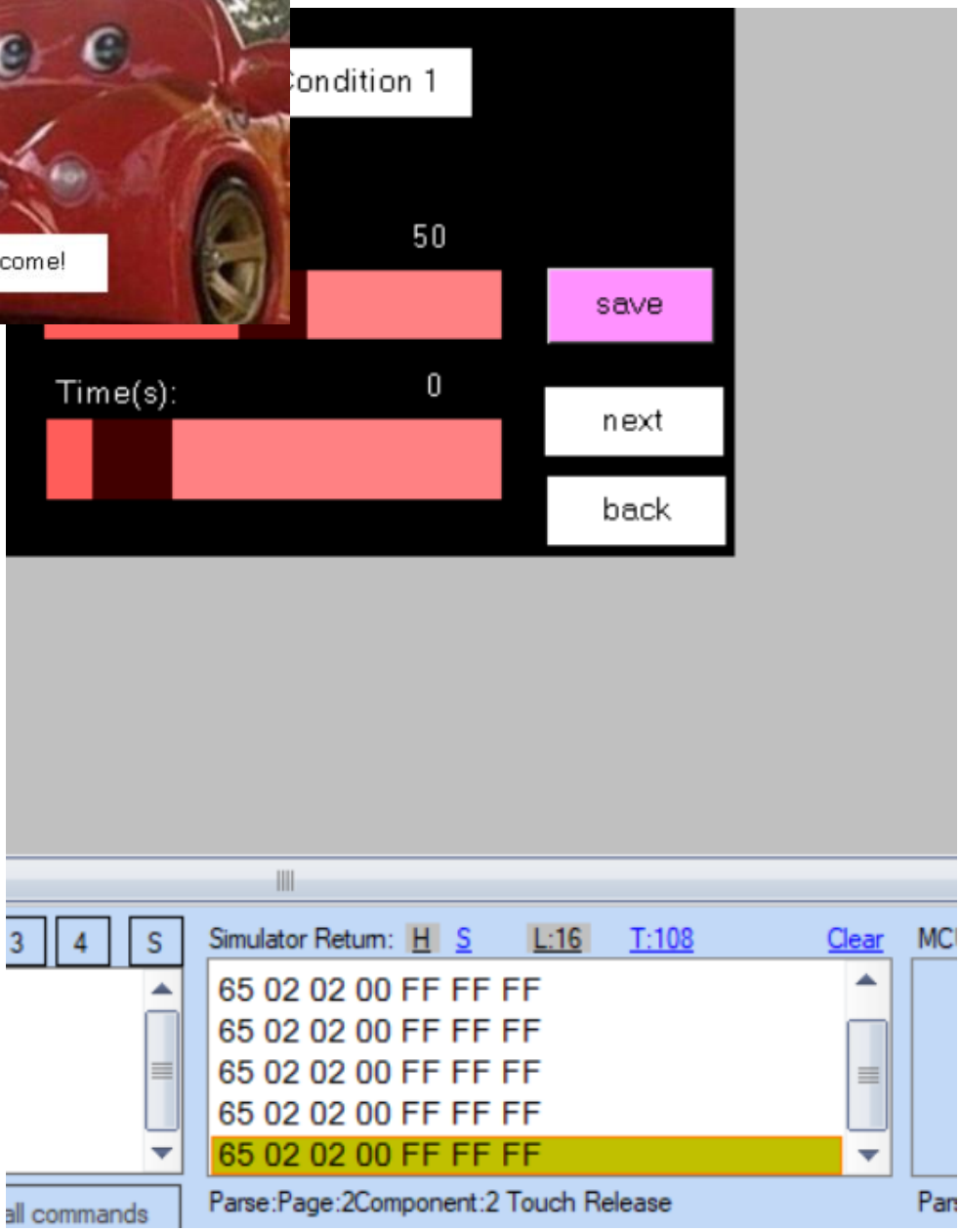
At first, I only did the first tasks to ensure that everything is working properly, which it did. I was able to change the values to whatever I wanted and I also added a slide-bar to see how it works since an older student suggested that it is easier to work with a slide-bar for the time and distance conditions of the car. Unfortunately I did not take any pictures to upload here and show my progress.

After being given our semester project tasks I quickly started on programming the display and making the pages that we needed in order to meet the requirements.

I will show a couple of pages:



Page 1: (*After pressing the welcome button you will be redirected to* **Page 2**)



Page 3:

First problem I have ran into here is the time not changing while sliding the slide-bar. Still couldn't figure out why.

Conclusion:

In summary of this task I did enjoy coding, it helped me understand systems better and how every electronic made has lines of code. Unfortunately not being able to fully connect the Arduino to the Nextion Display was a huge problem that I couldn't fix in time and neither could my group help me.